MIL-F-60991B (MU)
16 November 1972
SUPERSEDING
MIL-F-60991A (MU)
16 March 1970

MILITARY SPECIFICATION

FUZE, PD, M533

METAL PARTS FOR

AND

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the metal parts and loading, assembling and packing for one type of fuze designated as Fuze, PD, M533.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-60940	- Lead Cup Assembly

STANDARDS

FEDERAL

FED-STD-151	- Metals; Test Methods

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1390

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MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. Army Munitions Command

8876192	- Fuze, PD, M533

PUBLICATIONS

ARMY

ET-8876192	- Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents from a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

. Title 49	- Transportation, Parts -190

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)".)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Parts.-The parts shall comply with all requirements specified on Drawing (Dwg.) 8876192, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Qualification test for M55 Detonator.-The detonator shall function the test cartridge high order.

3.4 X-ray examination of sample fuze assemblies prior to jolt and jumble tests.-The fuze assembly shall be x-rayed and shall not be armed or partially armed (as evidenced by the detonator moving any degree from the safe position or presence of the set-back pin flange seated against bottom surface of the rotor plate (Dwg. 8876197) counterbore).

3.5 Transportation-vibration.-The fuze assembly assembled to test cartridge (Dwg. 9206914), and linked shall be safe and operable.

3.6 Jolt and Jumble.-The fuze assembly shall withstand the jolt and jumble tests without having any explosive element explode, fuze come apart, fuze arm or become hazardous to anyone handling them.

3.7 Waterproofness.-A continuous stream of air bubbles shall not emerge from the fuze assembly.

3.8 Functioning.-The fuze assembly shall function satisfactorily (See 6.6).

3.8.1 Safety.-There shall be no premature burst (See 6.7) or evidence thereof in any of the test firings.

3.9 Non-function.-The fuze assembly shall not function.

3.10 Workmanship.-All parts and assemblies shall be fabricated, finished, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

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3.11 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.5).

3.12 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

3.13 Contamination and distortion of Lexan parts.-Extreme care should be taken throughout all phases of manufacturer to avoid contaminating "Lexan" material. Specific information on "Lexan" distructive solvents may be obtained from the General Electric Co., Chemical and Metallurgical Division, Pittsfield, Mass.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unlesss otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

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d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.-

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

a. Three-hundred and thirty-four (334) fuze assemblies.
b. One-hundred and ten (110) sub-assemblies.
c. Fifty (50) each of every component.

All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process (including heat treatment procedures and equipment), material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspection to be performed.-All parts and assemblies shall be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

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4.2.2.1 Quantities for first article inspection.-After classification of defects and other inspections have been performed as specified in 4.2.2., the following quantities shall be subjected to this test:

4.2.2.1.1 Qualitative spot test for tin.-Five (5) rotor springs and ten (10) lock springs shall be subjected to this test.

4.2.2.1.2 Qualitative spot test for cadmium (when applicable).-Five (5) rotor springs and ten (10) lock springs shall be subjected to this test.

4.2.2.1.3 Stake of pinion assembly.-The one-hundred and ten (110) pinion assemblies shall be subjected to this test.

4.2.2.1.4 Push test of verge assembly.-The one-hundred and ten (110) verge assemblies shall be subjected to this test.

4.2.2.1.5 Stake of firing pin and spring assembly.-The one-hundred and ten (110) firing pin and spring assemblies shall be subjected to this test.

4.2.2.1.6 Force test of bracket pin of centerplate assembly.-Thirty-seven (37) centerplate assemblies and the bracket pin on each of three legs of each assembly shall be subjected to this test.

4.2.2.1.7 Functioning of centerplate assembly.-The one-hundred and ten (110) centerplate assemblies shall be subjected to this test.

4.2.2.1.8 Arming delay time and non-arming of housing assembly.-The one-hundred and ten (110) housing assemblies shall be subjected to this test.

4.2.2.1.9 Non-arming centrifugal lock of housing assembly.-The one-hundred and ten (110) housing assemblies shall be subjected to this test.

4.2.2.1.10 Arming of centrifugal lock of housing assembly.-The one-hundred and ten (110) housing assemblies shall be subjected to this test.

4.2.2.1.11 Push test of simulated setback pin into retainer setback pin.-The fifty (50) retainer setback pin shall be subjected to this test.

4.2.2.1.12 Pull test of retainer out of setback pin and housing assembly.-The one-hunrded and ten (110) setback pin and housing assemblies shall be subjected to this test.

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4.2.2.1.13 Spin test of housing assembly.-The one-hundred and ten (110) housing assemblies shall be subjected to this test.

4.2.2.1.14 X-ray examination of sample fuze assemblies, prior to jolt and jumble tests.-Prior to subjecting the fuze assemblies to the tests specified in 4.2.2.1.17 and 4.2.2.1.18, the fuze assemblies shall be subjected to this test.

4.2.2.1.15 Deflection test of centerplate.-The fifty (50) centerplates shall be subjected to this test.

4.2.2.1.16 Hardness of verge shaft and pinion shaft.-Fifty verge shafts and fifty (50) pinion shafts shall be subjected to this test.

4.2.2.1.17 Jolt.-Twelve (12) fuze assemblies shall be subjected to this test.

4.2.2.1.18 Jumble-Twelve (12) fuze assemblies shall be subjected to this test.

4.2.2.1.19 Gap measurement of bracket on centerplate assembly.-Forty-two (42) centerplate assemblies and the bracket on each of the three (3) legs of each assembly shall be subjected to this measurement.

4.2.2.1.20 Security of bracket on centerplate assembly.-If any bracket fails the gap requirement specified in 4.2.2.1.19, the forty-two (42) centerplate assemblies and the bracket on each of the three (3) legs of each assembly shall be subjected to this test.

4.2.3 Rejection.-If any part or assembly fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a part or assembly in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2.4 Remaining first article inspection samples.-

4.2.4.1 Transportation-vibration.-Three-hundred (300) fuzes assemblies, assembled to test cartridges (Dwg. 9206914), linked and packed in regular complete round shipping pack, shall be subjected to this test.

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4.2.4.2 Waterproofness.-After completion of test specified in 4.2.4.1, seventy-five (75) fuze assemblies shall be subjected to this test. If two (2) or more fuzes assemblies fail to comply with the applicable requirement, the first article sample shall be rejected.

4.2.4.3 Functioning.-After completion of tests specified in 4.2.4.1 and 4.2.4.2, the three-hundred (300) fuze assemblies shall be subjected to this test. If eight (8) or more fuze assemblies fail to comply with the applicable requirements, or one (1) or more premature burst occurs, the first article sample shall be rejected.

4.2.4.4 Non-functioning.-Ten (10) fuze assemblies shall be subjected to this test. If any fuze assembly functions, the first article sample shall be rejected.

4.2.5 Rejection.-The Government reserves the right to terminate its inspection upon any failure of a part or assembly in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.-

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a reprsentative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accor-

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dance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance MIL-STD-1168. If production break occurs more than 30 days, then when new production starts, the first 5 lots shall be limited to 35,000 units. Each inspection lot shall contain:

a. Detonators from one interfix lot number from one manufacturer.
b. Lead cup assemblies from one interfix lot number from one manufacturer.
c. A serial lot of fuzes may contain accepted ogives from more than one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Pin, firing (See Dwg. 8876228 covering a detail of dwg. 8876230).-

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Min. flat of firing pin point	Gage	01001
Minor:	AQL 0.6 Percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	01002

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4.3.2.2 Pin, setback (See dwg. 8876211 covering a detail of dwg. 8880916).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Diameter of shaft at end toward retainer	Gage	02001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	02002

4.3.2.3 Retainer, setback pin (See dwg. 8876212 covering a detail of dwg. 8880916).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Depth	Gage	03001
102.	Diameter, retaining mounds	Gage	03002
103.	Concentricity retaining mounds diameter with center diameter	Gage	03003
104.	Center diameter	Gage	03004
Minor:	AQL 1.00 Percent		
201.	Any retainer mounds missing	Visual	03005
202.	Evidence of poor workmanship (See 3.10)	Visual	03006

4.3.2.4 Ogive (See Dwg. 8876219 covering a detail of Dwg. 8876192).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Length from shoulder to end of node	Gage	04001
102.	Profile	Gage	04002

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103.	Concentricity of fuze cavity with large outside diameter including perpendicularity	Gage	04003
104.	Concentricity of pitch diameter of thread with large outside diameter	Gage	04004
105.	Diameter at end of probe	Gage	04010
106.	Length from bottom of fuze cavity to probe	Gage	04008
107.	Concentricity of diameter at end of probe with large outside diameter	Gage	04011

Minor: AQL 1.00 Percent

201.	Evidence of poor workmanship (See 3.10)	Visual	04005
202.	Length from shoulder to back end of thread	Gage	04006

4.3.2.5 Pin, push (See dwg. 8876223 covering a detail of dwg. 8876227).-

Categories	Defects	Method of Inspection	Code No.

Critical: None defined

Major: AQL 0.65 Percent

201.	Small diameter	Gage	05001
202.	Evidence of poor workmanship (See 3.10)	Visual	05002

4.3.2.6 Spring, bracket (See Dwg. 8876226).-

Categories	Defects	Method of Inspection	Code No.

Critical: None defined

Major AQL 0.40 Percent

101.	Load at deflected height	Test (See 4.4.27)	44001

Minor: None defined

4.3.2.7 Spring rotor (See Dwg. 8876215).-

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Load at assembled length	Test (See 4.4.28)	45001
Minor:	None defined		

4.3.2.8 Firing pin and spring assembly (See dwg. 8876230 covering a detail of dwg. 8876193).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Width of slot	Gage	06001
Minor:	None defined		

4.3.2.9 Setback pin and housing assembly (See dwg. 8880916 covering a detail of dwg. 8876193).-

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Rotor fails to remain in unarmed position when centrifugal lock is retracted	Visual/Manual	07001
2.	Setback pin not through rotor gear	Probe	07002
3.	Setback pin retainer missing	Visual/Probe	07003
4.	Two (2) or more leaves of setback pin retainer will not contact setback pin (See 4.3.2.15.1)	Visual	07004
Major:	AQL 0.40 Percent		
101.	One (1) leaf will not contact setback pin (See 4.3.2.15.1)	Visual	07005
102.	Discernible buldge on detonator	Visual	07006
Minor:	None defined		

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4.3.2.10 Rotor and gear assembly (See dwg. 8876202 covering a detail of dwg. 8876220).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Rotor, weight incorrect and coining missing	Visual	08001
Minor:	None defined		

4.3.2.11 Fuze assembly, prior to assembling lead cup assembly (See dwg. 8876192).-

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Setback pin retainer missing	Visual/Probe	09001
2.	Setback pin missing or re-traced (See 4.3.2.15.2)	Gage	09002
Major:	None defined		
Minor:	None defined		

4.3.2.12 Fuze assembly (See dwg. 8876192).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Pitch diameter of thread	Gage	10001
102.	Major diameter of thread	Gage	10002
103.	Large outside diameter, max.	Gage	10003
104.	Distance from end of ogive threaded end, to face of disc	Gage	10004
105.	Lead cup assembly missing	Visual	10005
106.	Crimp not fully 360 degrees	Visual	10006
Minor:	AQL 1.50 Percent		
201.	Diameter of thread undercut	Gage	10007
202.	Evidence of poor workmanship (See 3.10)	Visual	10008
203.	Color of ogive incorrect	Visual	10009

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4.3.2.13 Box, fiber, prior to sealing (See dwg. 8880918).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Incorrect number of fuzes	Visual	11001
Minor:	AQL 1.00 Percent		
201.	Pad missing on top of exposed area	Visual	11002
202.	Fillers missing	Visual	11003

4.3.2.14 Box, fiber (See dwg. 8880918).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Contents loose	Manual	12001
102.	Box damaged	Visual	12002
103.	Improper sealing	Visual	12003
Minor:	AQL 0.65 Percent		
201.	Marking misleading or unidentifiable	Visual	12004

4.3.2.15 Defects standards - Code No. (13001).-

4.3.2.15.1 Examination for contact of setback pin retainer leaves (See para. 4.3.2.9).-Each assembly in which one (1) or more leaves of the setback pin retainer apparently will fail to contact the setback pin shall be subjected to the following:

a. The setback pin shall be operated.

b. Visual examination shall be conducted to determine whether leaves contact the setback pin.

If two (2) or more leaves fail to contact the setback pin, the assembly shall be classed a critical defective. If one (1) leaf fails to contact the setback pin, the assembly shall be classed a Major defective. Assemblies subjected to examination in accordance with this procedure shall not be returned to the lot.

4.3.2.15.2 Examination for setback pin (See para. 4.3.2.11).- The position of the setback pin shall be classified a Critical defect if the top of the setback pin is less than .071 inches from the exposed surface of the lower shaft plate.

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4.3.3 Testing.-

4.3.3.1 Dimensional control of stamped parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.3.3.2 Check test for deterioration of detonators(See applicable detonator specification) - Major defect - Code No.(15001).- If the total time between original acceptance of any lot and the assembly of that lot into the fuze exceeds two years, or if the detonators have been subjected to adverse conditions, however brief, at any time since previous tests, the detonator lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in the applicable detonator specification immediately before the detonator lot is assembled into the fuze. This test shall be performed by the contractor on detonators selected by the Government inspector at the facility assembling the detonator into the fuze. (See 6.4)

4.3.3.3 Check test for deterioration of lead cup assemblies (See MIL-C-60940 - Major defect - Code No. (16001).-If the total time between original acceptance of any lot and the assembly of that lot into the fuze exceeds two years, or if the lead cup assemblies have been subjected to adverse conditions, however brief, at any time since previous tests, the lead cup lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in MIL-C-60940 immediately before the lead cup lot is assembled into the fuze. This test shall be performed by the contractor on lead cups selected by the Government inspector at the facility assembling the lead cup into the fuze. (See 6.4)

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4.3.3.4 Qualitative spot test for tin (See dwgs. 8876214 and 8876215).- Major defect - Code No. (17001).-Ten (10) lock springs and five (5) rotor springs shall be selected from each lot and subjected to this test. If any part fails to comply with the applicable requirement, the lot shall be rejected.

NOTE: For the purpose of this test, one lot of springs shall consist of parts manufactured from not more than one roll of spring wire.

This test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.5 Qualitative spot test for cadmium (See Dwg. 8876215 and 8876214), when applicable, Major defect, Code No. 37001.- Five (5) rotor springs and ten (10) lock springs shall be selected from each lot and subjected to this test. If any part fails to comply with the applicable requirement the lot shall be rejected.

NOTE: For the purpose of this test, one lot of springs shall consist of parts manufactured from not more than one roll of spring wire. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.6 Qualification test for M55 detonator (See Para. 3.3), Major defect, Code No. 36001.-One-hundred and twenty-five (125) detonators shall be randomly selected from each detonator lot and subjected to this test. If any detonator fails to comply with the applicable requirement, the detonator and lead cup combination shall be rejected for use in M533 fuzes and SMUPA-A-S shall be notified. NOTE: The detonators selected for this test shall be tested with the lead cup lot with which they will be used. Change in lead cup lot requires retest of detonators. The test shall be performed as specified in 4.4.3.

4.3.3.7 Dimensional inspection for "length to flange along smallest diameter minimum on pin setback (See Dwg. 8876211), Critical defect, Code No. 46001.-This inspection shall be conducted one-hundred percent. If any setback pin fails the dimensional requirement of the applicable drawing, it shall be classified defective and removed from the lot. The inspection shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.3.8 Deflection test of centerplate (See Dwg. 8876221), Major defect, Code No. 41001.-One-hundred and twenty-five (125) centerplate shall be randomly selected from each heat treat batch and subjected to this test. If two (2) or more centerplates fail to comply with the applicable requirement, the heat treat batch

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represented by the sample shall be rejected. The samples selected shall be representative of the heat treat batch. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4. (See 6.8).

4.3.3.9 Hardness of verge shaft and pinion shaft (See Dwg. 8876208), Major defect, Code No. 42001.-Twenty (20) of each verge shafts and pinion shafts shall be selected from each heat treat batch and subjected to this test. If one (1) or more of each twenty (20) fail to comply with the requirement specified on the applicable drawing, the batch represented by the sample shall be rejected. The test shall be performed as specified in 4.4.6.

4.3.3.10 Stake of pinion assembly (See Dwg. 8876210) - Major defect - Code No. (18001).-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. This test shall be performed as specified in 4.4.7 using equipment in accordance with 4.3.4.

4.3.3.11 Push Test of verge assembly (See dwg. 9256949) - Major defect - Code No. (19001).-The sample plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. This test shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4.

4.3.3.12 Stake of firing pin and spring assembly (See Dwg. 8876230) - Major defect - Code No. (20001).-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. This test shall be performed as specified in 4.4.9 using equipment in accordance with 4.3.4.

4.3.3.13 Force test of bracket pin of centerplate assembly (See dwg. 8876277) - Major defect - Code No. (21001).-Forty-two (42) centerplate assemblies shall be selected from each lot and the bracket pin on each of the three legs of each assembly shall be subjected to this test. If two (2) or more bracket pins fail to comply with the requirement specified on the applicable drawings, the lot shall be rejected. This test shall be performed as specified in 4.4.10 using equipment in accordance with 4.3.4.

4.3.3.14 Functioning of centerplate assembly (See dwg. 8876227)- Major defect - Code No. (22001).-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. This test shall be performed as specified in 4.4.11 using equipment in accordance with 4.3.4.

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4.3.3.15 Arming delay time (See Dwg. 8876220) - Critical defect if assembly arms in less than 100 milliseconds, Major defect if assembly arms between 100 and 140 milliseconds) - Non-arming of housing assembly - Major defect - Code No. (23001).- This test shall be conducted 100 percent. If any housing assembly fails to comply with the time limits specified on the applicable drawing, it shall be classified defective and removed from the lot. This test shall be performed as specified in 4.4.12 using equipment in accordance with 4.3.4.

4.3.3.16 Push test of simulated setback pin into retainer, setback pin (See dwg. 8876212) - Major defect - Code No. (24001).- The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. This test shall be performed as specified in 4.4.13 using equipment in accordance with 4.3.4.

4.3.3.17 Non-arming of centrifugal lock of housing assembly (See dwg. 8876220) - Major defect - Code No. (25001).-This test shall be conducted 100 percent. If any housing assembly fails to comply with the requirement specified on the applicable drawing, it shall be classed defective and removed from the lot. This test shall be performed as specified in 4.4.14 using equipment in accordance with 4.3.4.

4.3.3.18 Arming of centrifugal lock of housing assembly (See dwg. 8876220) - Major defect - Code No. (26001).-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter L, single plan, with an AQL of 0.40 percent. This test shall be conducted after completion of test specified in 4.3.3.17. This test shall be performed as specified in 4.4.15 using equipment specified in 4.3.4.

4.3.3.19 Transportation Vibration test of centrifugal lock mechanism (See dwg. 8876220) - Major defect - Code No. (27001).- Twenty (20) unarmed fuze assemblies without setback pins and without detonators, shall be selected from each lot and subjected to this test. Each fuze assembly cover shall be conspicuously marked to indicate absence of setback pin and detonator. Fuze assemblies shall be placed in interplant shipping containers and subjected to this test.

At the outset of the test, the rotor shall be in the fully unarmed position, with the rotor gear in contact with a pin with a diameter of .078 - .001 inches which passes through the rotor gear and rotor plate. The centrifugal lock shall be in engagement with the star wheel. The pin shall then be removed and the fuze

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assembly shall be subjected to the test. The amount of travel of each rotor shall be noted. If the travel of any rotor exceeds 26 degrees the lot shall be rejected. All data concerning rotor travel shall be reported to Picatinny Arsenal, ATTN: SMUPA-QA-A-S. This test shall be performed as specified in 4.4.16 using equipment specified in 4.3.4.

4.3.3.20 Gap measurement of bracket on centerplate assembly (See dwg. 8876227) - Major defect - Code No. (28001).-Forty-two (42) centerplate assemblies shall be selected from each lot and bracket on each of the three (3) legs of each assembly shall be subjected to this measurement. If less than two (2) brackets fail to comply with the requirement specified on the applicable drawing, the lot shall be accepted. If two (2) or more brackets fail to comply with the requirement specified on the applicable drawing, the test specified in Paragraph 4.3.3.21 shall be performed. This test shall be performed as specified in 4.4.17 using equipment in accordance with 4.3.4.

4.3.3.21 Security of bracket on centerplate assembly (See dwg. 8876227) - Major defect - Code No. (29001).-If two (2) or more brackets fail the test specified in 4.3.3.20, the forty-two (42) centerplate assemblies shall be subjected to this test. If two (2) or more brackets fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.18 using equipment in accordance with 4.3.4.

4.3.3.22 X-ray examination of sample fuze assemblies prior to jolt and jumble test,, (See 3.13), Critical defect, Code No. (38001).-The samples selected for the test specified in 4.3.3.23 and 4.3.3.24 shall be subjected to this test. If any assembly fails the requirement of 3.4, the lot shall be rejected. This test shall be performed as specified in 4.4.19.

4.3.3.23 Jolt (See para. 3.6) - Major defect - Code No. (30001).- Twelve (12) fuze assemblies shall be selected from each lot and subjected to this test. If any fuze assembly fails to comply with the applicable requirement, the lot shall be rejected. This test shall be performed as specified in 4.4.20 using equipment specified in 4.3.4.

4.3.3.24 Jumble (See para. 3.6) - Major defect - Code No. (31001).-Ninety-six (96) fuze assemblies shall be selected from each lot and subjected to this test. If any fuze assembly fails to comply with the applicable requirement, the lot shall be rejected. This test shall be performed as specified in 4.4.20 using equipment specified in 4.3.4.

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4.3.3.25 Transportation vibration (See para. (3.5), Major defect, Code No. 32001.-Two-hundred and ten (210) fuze assemblies assembled to test cartridges (Dwg. 9206914), linked, placed in regular complete round shipping pack, and subjected to this test. The test shall be performed as specified in 4.4.23 using equipment specified in 4.3.4.

4.3.3.26 Pull test of retainer out of setback pin and housing assembly (See dwg. 8880916), Major defect, Code No. 48001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. Applies to one (1) pound minimum requirement only. The test shall be performed as specified in 4.4.21 using equipment in accordance with 4.3.4.

4.3.3.27 Spin test of housing assembly (See dwg. 8876220), Major defect, Code No. 47001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.65 percent. The test shall be performed as specified in 4.4.22 using equipment in accordance with 4.3.4.

4.3.3.28 Waterproofness (See para. 3.7) - Major defect - Code No. (33001).-After completion of test specified in 4.3.3.25, eighty (80) of the sample fuze assemblies shall be subjected to this test. If four (4) or more fuze assemblies fail to comply with the applicable requirement, the lot shall be rejected. Assemblies which failed this test shall be discarded and replaced with units which had been subjected to test specified in 4.3.3.25, prior to subsequent testing. This test shall be performed as specified in 4.4.24 using equipment specified in 4.3.4.

4.3.3.29 Functioning (See para. 3.8) - Major defect - Code No. (34001).-After completion of tests specified in 4.3.3.25 and 4.3.3.28, two hundred (200) fuze assemblies shall be subjected to this test. If eight (8) or more fuze assemblies fail to comply with the applicable requirement, or one (1) or more premature burst occurs the lot shall be rejected. This test shall be performed as specified in 4.4.25 using equipment specified in 4.3.4.

4.3.3.30 Non-functioning (See para. 3.9) - Major defect - Code No. (35001).-Ten (10) additional fuze assemblies shall be subjected to this test. If any fuze assembly functions, the lot shall be rejected. This test shall be performed as specified in 4.4.26 using equipment specified in 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 8876192 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to performed the examinations and

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and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Qualitative spot test for tin.-The rotor springs and lock springs shall be tested in accordance with the following procedure:

a. Prepare 50 ml. of saturated solution of cacotheline by dissolving small quantities of cacotheline powder in water. The solution can be considered saturated when the addition of small amounts of cacotheline will no longer dissolve.

b. Clean the parts in acetone and place 2 lock springs or one (1) rotor spring in a clean test tube. Add 10 drops of concentrated hydrochloric acid and allow tin to dissolve for 10 minutes. Then add 5 drops of distilled water and mix well.

c. Moisten No. 42 Whatman filter paper with saturated cacotheline solution and allow to partially dry.

d. Place a drop of test solution from the test tube on damp filter paper. The development of a distinct red-violet spot where the test solution was placed indicates the presence of tin. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.2 Qualitative spot test for cadmium.-The rotor springs and lock springs shall be tested in accordance with the following procedure:

a. Clean the parts in acetone and place one (1) rotor spring or two (2) lock springs in a clean test tube. Add 1.0 ml. concentrated hydrochloric acid and allow cadmium to dissolve.

b. Decant fluid into a breaker and dilute with distilled water to total 100 ml.

c. Add 50 ml. saturated hydrogen sulfide.

d. The development of a yellow precipitant indicates the presence of cadmium.

This test is a destructive test. Parts so tested shall not be returned to the lot.

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4.4.3 Qualification test for M55 detonator.-The detonator and lead cups shall be assembled to a M533 fuze and assembled to a test cartridge (dwg. 9231745). The test cartridge shall be conditioned at -65°F for 16 hours min. and fired in accordance with paragraph 4.4.23.

4.4.4 Dimensional inspection for "length to flange along smallest diameter minimum" on pin setback.-The setback pin shall be gaged using the equipment specified on the Equipment List. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.5 Deflection test of centerplate.-The centerplate shall be placed in the fixture specified on the Equipment List and the force specified on dwg. 8876221 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.6 Hardness of verge shaft and pinion shaft.-The verge shaft and pinion shaft shall be subjected to the Rockwell hardness test in accordance with ASTM test Method E 18-67. Parts so tested shall not be returned to the lot.

4.4.7 Stake of pinion assembly.-The pinion assembly shall be placed in a fixtrue specified on the Equipment List and the torque specified on dwg. 8876210 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.8 Push test of verge assembly.-The verge assembly shall be placed in a fixture specified on the Equipment List and the load specified on dwg. 9256949 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.9 Stake of firing pin and spring assembly.-The firing pin and spring assembly shall be placed in a fixture specified on the Equipment List and the minimum twist specified on dwg. 8876230 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.10 Force test of bracket pin of centerplate assembly.-The centerplate assembly shall be placed in a fixture specified on the Equipment List and the force specified on dwg. 8876226 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

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4.4.11 Functioning of centerplate assembly.-The centerplate assembly shall be placed in a holder specified on the Equipment List. The ring shall be depressed to contact the centerplate and freedom of movement of the brackets and push pin shall be observed. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.12 Arming delay time and non-arming of housing assembly.- An electronic time device and an electrically operated retractable simulated setback pin shall be operated simultaneously. A light source shall be concentrated on the detonator hole in the rotor. A photoelectric cell shall not trip an electronic circuit before the non-arming time specified. For the arming time, the photoelectric cell shall trip an electric circuit by the arming time specified. Other approved methods may be used. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.13 Push test of simulated setback pin into retainer setback pin.-The retainer setback pin shall be placed in a fixture as shown on the Equipment List and the force specified on dwg. 8876212 shall be applied to a simulated mean setback pin. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.14 Non-arming of centrifugal lock of housing assembly Method 1.-The housing assembly shall be mounted in a centrifuge and spun about its axis of symmetry at the rotational speed specified for a minimum of 0.400 seconds. Visually check to see that the rotor has not moved from the unarmed position. When the visual check provides doubtful results, a gage shall be used which simulated the minimum diameter set back pin. An unarmed housing assembly shall be capable of accepting the simulated pin through the rotor plate and rotor gear.

Method 2.-The housing shall be mounted and spun as in Method 1. Upon completion of spin, a light source shall be concentrated on the rotor through the setback pin check hole in the top plate. A photoelectric cell shall trip an electric circuit. Other approved methods may be used. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.15 Arming of centrifugal lock of housing assembly, Method 1.- The housing assembly shall be mounted in a centrifuge and spun about its axis of symmetry at the rotational speed specified. Visually check to see that the rotor moves from the unarmed position to the armed position.

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Method 2.-The housing shall be mounted and spun as in Method 1. Upon completion of spin, a light source shall be concentrated on the rotor through the setback pin check hole in the top plate. A photoelectric cell shall trip an electric circuit. Other approved methods may be used. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.16 Transportation vibration test of centrifugal lock mechanism.-The mechanism shall be tested as specified in MIL-STD-331. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.17 Gap measurement of bracket on centerplate assembly.-The gap measurement shall be gaged as shown on the Equipment List. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.18 Security of bracket on centerplate assembly.-The centerplate assembly shall be placed in the fixture specified on the Equipment List and the force specified on dwg. 8876227 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.19 X-ray examination of sample fuze assemblies prior to jolt and jumble tests.-Armed or partially armed fuze assemblies shall be determined by X-ray equipment or any other method satisfactory to the Contracting Officer. This test shall be conducted in two (2) phases: (a) From the top of the fuze assembly down and (b) from the side. Side view shall show the flange of the setback pin seated against bottom surface of the rotor plate (Dwg. 8876197) counterbore. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.20 Jolt and Jumble.-The fuze assemblies shall be tested in accordance with the procedures specified in MIL-STD-331. These tests are destructive tests. Parts so tested shall not be returned to the lot.

4.4.21 Pull test of setback pin and housing assembly.-The setback pin and housing assembly shall be placed in the fixture specified on the Equipment List and the pull specified on dwg. 8880916 shall be applied to the retainer. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.22 Spin test of housing assembly.-The housing assembly shall be mounted in a centrifuge spin fixture and rotated at 14,000 RPM., as specified on Dwg. 8876220. Observation shall be

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be made that the rotor and gear assembly arm in accordance with the requirement specified on Dwg. 8876220. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.23 Transportation vibration.-The fuze assemblies, assembled to test cartridges (Dwg. 9206914), linked, shall be tested in accordance with MIL-STD-331. Parts subjected to this test shall not be returned to the lot but may be used for subsequent tests.

4.4.24 Waterproofness.-The fuze assembly shall be immersed in water to a depth of not less than 3 inches. Both fuze assembly and water shall be at ambient temperature at start of test. The chamber shall be subjected to a vacuum pressure differential of 5 p.s.i. for a period of not less than 15 seconds. The water shall contain the wetting agent Nacconol NR (sodium alkyl aryl sulfonate) or equal in a concentration of approximately 1 1/2 teaspoonsful per gallon of water. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.25 Functioning.-The fuze assembly shall be assembled to a test cartridge (dwg. 9206914) and fired from a launcher for which it is standard to impact at 200 meters plus or minus 5 meters onto coarse sand, ten inches minimum thickness at point of impact. The angle of fring shall be adjusted to give an impact of approximately 30 degrees. Rake sand after every round or traverse weapon between shots so that impacts are not overlapping. If a round falls outsidethe specified area, it shall be declared a "no test" round. When results for functioning are reported, the number of "no test" rounds shall also be reported. This test is destructive.

4.4.26 Non-functioning.-The fuze assembly shall be assembled to a test cartridge (dwg. 9206914) and fired from a launcher for which it is standard against a vertical colotex target placed 60 feet from the launcher. Observation shall be made for non-functioning. This test is destructive.

4.4.27 Load at deflected height of spring, bracket.-The spring, bracket shall be placed in the fixture specified on the Equipment List and the load specified on Dwg. 8876226 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.28 Load at assembled length of spring, rotor.-The spring rotor shall be placed in the fixture specified on the Equipment List and the load specified on Dwg. 8876215 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level C.-The fuzes shall be preserved and packaged in accordance with dwg. 8880918.

5.2 Packing.-

5.2.1 Level C.-The fuzes, preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with dwg. 9230368.

5.3 Marking.-Marking shall be in accordance with Dwg. 8799112 and Code of Federal Regulations, Title 49 Parts 0-190.

5.4 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.
d. Provision for check testing of primer and detonator. (See 6.4)

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collections and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (See 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010)

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the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (See 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to Specification MIL-D-1000, Category E, Form 2. Legibility and

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reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Cost of check test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the detonator or lead cup assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the detonator or lead cup assemblies or to the contractor assembling the detonator or lead cup assemblies into the fuze, and shall not constitute a basis for rejection against either contractor except where deterioration has occured as a direct result of carelessness in handling, storage, etc., permitted while the detonator or lead cup assembly lot was under the jurisdiction of either contractor.

6.5 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S.

6.6 Satisfactory functioning is defined as high order detonation evidenced by detonation of the lead cup and puncture of the disc in excess of 3/16" diameter.

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6.7 Premature burst.-A premature burst is considered to have occurred if the round functions in the gun bore or in flight.

6.8 If the contractor uses a continuous heat treat process, an equivalent sampling plan per MIL-STD-1235 may be used if approved by the procuring activity.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1390-A-007

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